

Streetrack Ubuntu 24.04 STIG Compliance Pipeline

Residual Findings Summary

Purpose. This deliverable summarizes the 16 STIG controls that remained failed after the final verification assessment. The findings are grouped into practical categories so the project can explain residual risk without creating a control-by-control remediation matrix.

Project evidence reviewed. Streetrack Ubuntu STIG baseline Tenable report dated August 23, 2026, and Streetrack Ubuntu STIG verification Tenable report dated August 23, 2026.

1. Verification Result

Baseline: 127 failed controls, 55 passed, 12 warning/manual review, 30.2% measured compliance.

Verification: 16 failed controls, 165 passed, 13 warning/manual review, 91.2% measured compliance.

Improvement: 111 fewer failed controls, approximately 87.4% reduction in failures, and a 60.9 percentage-point compliance increase.

2. Residual Finding Categories

Category	Controls	Count
Enterprise identity, PKI, and smart-card dependencies	UBTU-24-100650, UBTU-24-100660, UBTU-24-400020, UBTU-24-400030, UBTU-24-400060, UBTU-24-400340, UBTU-24-400360, UBTU-24-400370, UBTU-24-400375, UBTU-24-400380	10
External or centralized infrastructure dependency	UBTU-24-100450	1
Platform-level cryptographic requirement	UBTU-24-600030	1
Boot and time-environment limitations	UBTU-24-102000, UBTU-24-600160	2
Remaining local configuration weaknesses	UBTU-24-600070, UBTU-24-700020	2

Total residual failed controls: 16

2.1 Enterprise identity, PKI, and smart-card dependencies

Controls: UBTU-24-100650, UBTU-24-100660, UBTU-24-400020, UBTU-24-400030, UBTU-24-400060, UBTU-24-400340, UBTU-24-400360, UBTU-24-400370, UBTU-24-400375, UBTU-24-400380

These controls depend on enterprise identity infrastructure such as SSSD, smart-card/PIV authentication, PKI trust chains, certificate mapping, and revocation services. That supporting infrastructure was outside the scope of the single-server lab implementation.

2.2 External or centralized infrastructure dependency

Controls: UBTU-24-100450

This control requires audit records to be offloaded to a separate system or storage location. The project did not deploy a centralized audit collection server, so full remediation would require infrastructure beyond the defined capstone scope.

2.3 Platform-level cryptographic requirement

Controls: UBTU-24-600030

This control requires systemwide FIPS-validated cryptography. The project applied STIG hardening but did not convert the Ubuntu platform into a fully FIPS-enabled environment because that would introduce a broader platform and support requirement.

2.4 Boot and time-environment limitations

Controls: UBTU-24-102000, UBTU-24-600160

These controls concern authentication during maintenance/single-user boot and validation against an authoritative time source. They remain environment-specific limitations within the Azure lab context.

2.5 Remaining local configuration weaknesses

Controls: UBTU-24-600070, UBTU-24-700020

These controls are locally applicable settings that were not fully remediated by the implemented Ansible roles. They are retained as residual weaknesses and future remediation opportunities rather than expanding the current project scope.

3. Residual Risk Interpretation

The verification results show that most remaining failures are not evidence that the overall hardening workflow failed. Ten of the 16 residual failures depend on enterprise identity, PKI, or smart-card capabilities that were not available in the project environment. Additional findings depend on centralized logging, platform-level FIPS operation, or environment-specific boot and time services. Two controls remain as local configuration weaknesses. No verification-detected regression was present in the final assessment.

These residual findings should therefore be treated as documented limitations and future remediation opportunities. They do not negate the measured improvement from 30.2% to 91.2% STIG compliance or the reduction from 127 to 16 failed controls.

4. Recommended Treatment

- Accept enterprise identity, PKI, smart-card, and centralized-service findings as out of scope for the current lab implementation.
- Document the FIPS requirement as a platform-level limitation requiring separate planning and validation.
- Retain the two local configuration findings as future remediation candidates.
- Reassess residual findings when the STIG version, Ubuntu version, Ansible roles, or supporting infrastructure changes.